

Question	Answer	Marks	Guidance
4(a)	collisions are instantaneous	B1	<u>List Rules apply.</u> Allow duration of collisions is negligible compared with time between collisions.
	collisions are elastic	B1	Ignore references to conservation of energy.
4(b)(i)	change in momentum = $2mu$	A1	Ignore sign of answer.
4(b)(ii)	time = $2L / u$	A1	
4(b)(iii)	mean force = $\Delta p / t$ = $2mu / (2L / u)$ = mu^2 / L	A1	Ignore sign of answer.
4(b)(iv)	clear application of Newton's third law (e.g. force exerted on face by particle equal (and opposite) to force exerted on particle by face)	B1	
	$p = F / A$ = $(mu^2 / L) / L^2$ = mu^2 / L^3	A1	Ignore sign of answer.

Question	Answer	Marks	Guidance
4(c)	volume of cube is L^3 (which is V , so equation becomes $pV = mu^2$)	B1	Any explicit equating of V with L^3 is sufficient.
	particles moving in 3 dimensions, so u^2 becomes $\frac{1}{3}\langle c^2 \rangle$	B1	Not just algebra, need to see the explanation. Allow 'all' for '3'. Allow 'directions' for 'dimensions'. Allow ' c_x ' for ' u '. Allow ' $\langle c \rangle^2$ ' for ' $\langle c^2 \rangle$ ' for this marking point.
	N particles (rather than 1 particle) so (pV) multiplied by N or multiply by N to get pV for all particles (rather than just 1 particle)	B1	Not just algebra, need to see the explanation. May be seen as an expression for p or pV without N followed by the same expression with N , with reasoning that this is because there are N particles.